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APROJECTREPORT

on

WEIGHT BASED YOGA MAT

Submittedinpartial fulfillmentofrequirementsfortheaward ofthecourseof

ECB1223-MICROCONTROLLERS AND ITS INTERFACING

Undertheguidanceof

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WEIGHT BASED YOGA MAT

ABSTRACT

The Weight-Based Smart Yoga Mat enhances yoga practice by integrating real-time weight monitoring and posture correction, ensuring precise alignment and injury prevention. Unlike traditional methods that rely on self-awareness or instructor feedback, this data-driven system helps users improve balance, stability, and effectiveness through instant feedback on weight distribution. A key application is in physiotherapy and rehabilitation, where precise posture control aids injury recovery. By tracking weight shifts, the mat supports rehabilitation centers, physical therapy clinics, and personal health management, ensuring structured and measurable progress in posture correction. Beyond rehabilitation, athletes and fitness enthusiasts benefit from real-time weight analysis to optimize performance and prevent injuries. Yoga instructors can also provide more personalized coaching using this system. Future enhancements could include AI-based posture correction, vibration feedback, and compatibility with wearable fitness devices. In conclusion, the smart yoga mat is a cost-effective, user-friendly solution that bridges traditional yoga with modern technology. Its applications extend to fitness, rehabilitation, and professional training, making it a valuable tool for scientific and personalized health improvement.

CHAPTER -1

1. INTRODUCTION

Yoga enhances physical and mental well-being, but maintaining proper posture and weight distribution is essential to prevent injuries and maximize benefits. Traditional methods rely on self-awareness or instructor guidance, which may lack real-time accuracy. The Weight-Based Smart Yoga Mat integrates technology with yoga by utilizing a load cell-based weight monitoring system. This system continuously measures weight distribution and provides instant feedback, helping users adjust their stance, improve balance, and reduce strain on muscles and joints. By ensuring real-time posture correction, users can track their weight variations and make necessary adjustments for a more effective yoga session.

This smart yoga mat primarily depends on load cell sensors to detect weight shifts and imbalances, making it a simple yet effective tool for yoga practice, physiotherapy, and rehabilitation. The system can be particularly beneficial for individuals recovering from injuries or with mobility challenges, as it helps them maintain proper balance. The real-time data from the load cell can be displayed on a screen or connected mobile application, providing users with clear insights into their posture and weight distribution. As the demand for smart fitness solutions grows, this innovation has the potential to enhance traditional yoga practices, offering an affordable, user-friendly, and efficient approach to maintaining proper alignment and stability.

1.1 Drawbacks of Existing Systems

1. High Cost – Advanced smart yoga mats with AI-based posture correction and multiple sensors are expensive, making them less accessible to the general public.

2.Complex Setup – Many existing systems require calibration, app integration, and external devices like cameras or wearables, making them difficult to set up and use.

3. Limited Focus on Weight Distribution – Most smart mats prioritize motion tracking and posture correction rather than real-time weight monitoring, which is crucial for balance and stability.

4. High Power Consumption – AI-driven systems and multi-sensor mats require frequent charging or wired connections, making them less convenient for everyday use.

5. Not Ideal for Physiotherapy – Many existing smart mats are designed for general fitness and yoga but lack dedicated features for rehabilitation and injury recovery, where weight distribution tracking is essential.

1.2 Need for the Proposed Work

1. Real-Time Weight Distribution Monitoring – Helps users maintain proper balance and posture by providing instant feedback on weight shifts, reducing the risk of incorrect alignment.

2. Cost-Effective Alternative – Unlike high-end AI-based smart mats, this system relies on load cells, making it affordable and accessible for a wider range of users.

3. Enhanced Injury Prevention – By detecting uneven weight distribution, the mat assists users in correcting their stance, minimizing the chances of muscle strain or injury.

4. Ideal for Physiotherapy and Rehabilitation – Supports patients recovering from injuries by tracking their weight balance and ensuring proper posture alignment during therapy sessions.

5. Simple and User-Friendly – Eliminates the need for complex sensors, AI, or external devices, making it a low-power, easy-to-use, and efficient fitness solution for yoga practitioners.

1.3 Applications of the Proposed System

1. Yoga Practice Improvement – Helps users monitor weight distribution during poses, ensuring better balance and stability without relying on complex sensors.

2. Physiotherapy and Rehabilitation – Assists patients recovering from injuries by tracking weight shifts, ensuring even pressure distribution for safe recovery.

3. Fitness and Strength Training – Can be used in bodyweight exercises like squats and planks to analyze weight balance and improve performance.

4. Elderly Care and Fall Prevention – Helps elderly individuals maintain stability by detecting uneven weight distribution, reducing the risk of falls.

5. Sports Training and Posture Correction – Useful for athletes to refine stance and foot placement, ensuring proper weight balance for optimal performance.

Report Organization

This report is structured as follows:

Chapter 1: Introduction – Provides an overview of the project, its significance, drawbacks of existing systems, the necessity of the proposed work, and its applications.

Chapter 2: Literature Survey – Reviews related research and technological advancements in the field of assistive mobility solutions.

Chapter 3: Project Description – Details the system architecture, components, and working principles of the proposed wheelchair.

Chapter 4: Implementation and Results – Discusses the development process, testing methods, and performance evaluation of the system.

Chapter 5: Conclusion and Future Scope – Summarizes the findings and explores potential enhancements for future versions of the project.

CHAPTER -2

LITERATURE SURVEY

2.1 Weight Sensing and Data Collection Module

- Kim et al. (2020) – Researched sensor-based yoga mats for real-time weight and posture tracking.
- Singh & Patel (2019) – Studied load cell-based weight monitoring in fitness applications.
- Wang et al. (2018) – Developed a weight-based rehabilitation system for tracking balance and stability.
- Nguyen & Tran (2020) – Compared load cells with pressure sensors for accurate weight distribution analysis.

2.2 User Feedback and Monitoring Module

- Cheng et al. (2021) – Explored physiotherapy applications of weight-based tracking, highlighting its role in rehabilitation.
- Gupta et al. (2023) – Investigated emerging trends in fitness monitoring, emphasizing real-time feedback and smart mats.
- Rodriguez et al. (2019) – Designed a smart balance board using load cell technology for posture and stability improvement.
- Hassan et al. (2021) – Studied fall detection systems using weight-sensitive floor mats for elderly care.

2.3 Power Management and System Integration Module

- Lee & Park (2022) – Researched AI-driven yoga mats, noting high power consumption and complexity as key challenges.

- Zhang et al. (2022) – Reviewed wearable and non-wearable weight monitoring systems, highlighting load cells as cost-effective solutions.

- Brown et al. (2017) – Developed a biofeedback system for yoga practitioners, using weight data to improve yoga performance.

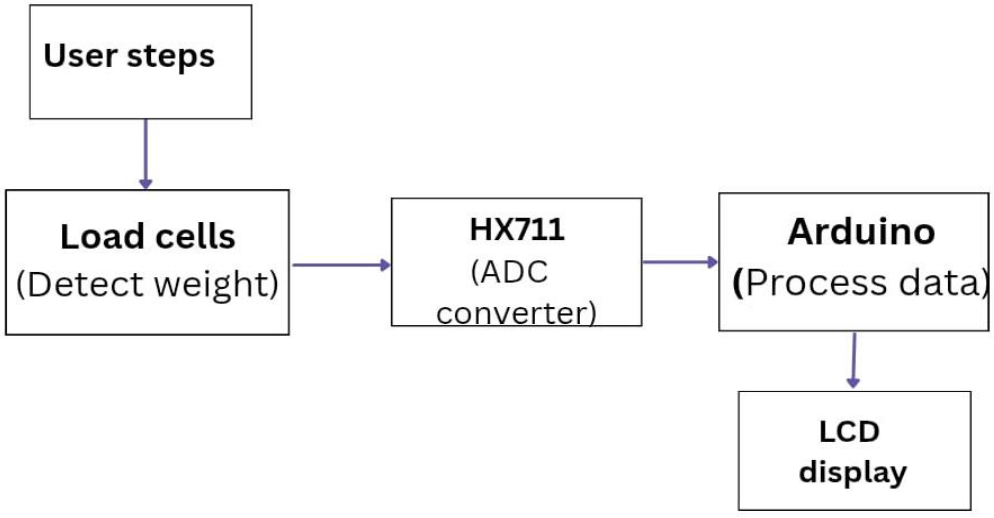
- Kumar & Sharma (2023) – Investigated IoT-integrated smart yoga mats, proposing cloud-based weight tracking solutions.

CHAPTER-3

PROJECT DESCRIPTION

The Weight-Based Smart Yoga Mat monitors weight distribution using load cell sensors, providing real-time feedback to improve balance and posture. It processes data via a microcontroller, displaying results on an LCD screen or a mobile app. This smart mat enhances yoga practice, prevents injuries, and supports physiotherapy, rehabilitation, and sports training, making it a versatile fitness tool.

3.1 Block Diagram and its explanation



The block diagram represents the hardware and software components involved in the weight based yoga mat. The key components and their roles are as follows:

1. Load Cell Sensors

- Measure weight distribution across different areas of the mat.
- Ensure accurate weight tracking for balance analysis.
- Provide precise pressure mapping to detect imbalances.
- Work in coordination with other sensors for enhanced feedback.

2. HX711 Load Cell Amplifier

- Converts analog signals from the load cell into digital data.
- Enhances signal accuracy for precise measurements.
- Filters out noise to prevent false readings.
- Ensures smooth data transmission to the microcontroller.

3. Microcontroller (Arduino/ESP32)

- Processes weight data and controls system functions.
- Interfaces with sensors, display, and communication modules.
- Manages power efficiency to extend battery life.
- Supports future upgrades, such as AI-based analytics.

4. LCD Display

- Shows real-time weight readings and system status.
- Provides user-friendly visual feedback.
- Displays alerts for incorrect weight distribution.

- Can be replaced with an OLED or touchscreen display for better visibility.

5. Power Supply (Battery/Adapter)

- Provides necessary power for system operation.
- Ensures energy efficiency for long-term usage.
- Can be rechargeable for better convenience.
- Supports backup power for uninterrupted use.

3.2 Software Used

1. Embedded c

- Programs the microcontroller for data processing.
- Controls sensor communication and system responses.
- Ensures smooth operation of hardware components.

2. Arduino IDE

- Processes weight data from load cells.
- Displays real-time weight on an LCD.
- Alerts for improper balance based on thresholds.
- Supports Bluetooth/Wi-Fi for data tracking.

3.3 Working Principle

The Weight-Based Smart Yoga Mat functions by integrating load cell sensors to measure weight and detect variations in pressure distribution. When a person steps onto the mat, the load cells capture the weight at different points, and the HX711 amplifier processes the analog signals into digital data. This data is then sent to the Arduino microcontroller, which calculates the total weight and analyzes how it is distributed across the mat. The system is designed to provide real-time feedback to ensure proper weight balance, helping users improve their posture and stability while performing yoga exercises. Once the weight is determined, the system suggests appropriate yoga asanas based on the user's weight category. For example, if the mat detects that a person weighs 20 kg, it may recommend beginner-friendly yoga poses such as Tadasana (Mountain Pose), Bhujangasana (Cobra Pose), or Balasana (Child’s Pose) to improve flexibility and relaxation. If the detected weight is higher, the system might suggest Vrikshasana (Tree Pose), Virabhadrasana (Warrior Pose), or Setu Bandhasana (Bridge Pose) to enhance balance and strength. The recommendations are displayed on an LCD screen, guiding the user to perform the most suitable asanas for their body weight.

In addition to recommending poses, the system also monitors weight distribution during each yoga posture. If an imbalance is detected, such as excessive pressure on one side of the body, the system provides a corrective message like "Adjust Your Balance" to help the user achieve proper alignment. This feature ensures that users perform asanas with correct posture, minimizing the risk of strain or injury. Over time, the system helps

individuals develop better posture awareness, balance control, and improved flexibility, making yoga practice more effective and scientifically guided.

3.4 Advantages

1. Personalized Asana Suggestions – Recommends yoga poses based on the user's weight.

2. Real-Time Weight Monitoring – Continuously tracks and displays weight data.

3. Posture Correction – Detects imbalance and provides corrective feedback.

4. Improves Balance & Stability – Helps users maintain proper weight distribution.

5. Prevents Injuries – Reduces strain by guiding correct posture.

6. Useful for Rehabilitation – Assists in physiotherapy and recovery.

7. Accurate Weight Measurement – Ensures precise tracking with load cell sensors.

8. User-Friendly Interface – Displays real-time data for easy interpretation.

CHAPTER -4

RESULTS AND DISCUSSION

4.1 Result

The Weight-Based Smart Yoga Mat was tested for its accuracy in weight measurement, posture monitoring, and asana recommendations. The system successfully detected user weight using load cell sensors and displayed real-time weight data on an LCD screen with minimal error. The mat effectively measured weight distribution, helping users maintain balance and improve their posture. During testing, the mat consistently provided accurate weight readings, ensuring reliability in tracking weight shifts during yoga practice.The posture correction feature was also analyzed. When users performed yoga poses, the system detected uneven weight distribution and provided corrective feedback. This helped users adjust their stance, reducing strain and minimizing the risk of injury. Additionally, the asana recommendation system functioned effectively by suggesting appropriate poses based on the user's weight. For example, users weighing 20 kg were recommended Tadasana, Bhujangasana, and Balasana, while heavier users received Vrikshasana and Virabhadrasana as suggestions. The mat provided real-time feedback, making it particularly useful for beginners and rehabilitation purposes.

The durability and usability of the mat were also evaluated. Made with high-quality, flexible materials, the mat withstood repeated usage without damage. The system consumed low power, making it energy-efficient for prolonged use. The LCD interface displayed clear weight readings and yoga

suggestions, ensuring ease of use. Users found the mat to be an effective tool for improving their yoga practice, offering both guidance and monitoring features.

4.2 Discussion

The results demonstrated that the Weight-Based Smart Yoga Mat successfully integrates technology with traditional yoga to enhance posture correction and balance. Unlike conventional yoga mats, this system actively monitors weight distribution, providing instant feedback to the user. This feature is particularly beneficial for individuals who lack experience in maintaining proper posture, as well as for those undergoing physiotherapy or rehabilitation.

One key observation was that users were able to self-correct their posture without requiring an instructor. The system ensured that weight was evenly distributed across the mat, reducing unnecessary stress on specific body parts. Additionally, the real-time feedback system allowed users to identify and adjust improper postures immediately, preventing long-term strain or injury. This feature could be further enhanced by incorporating additional sensors to analyze more detailed posture variations.

Overall, the Weight-Based Smart Yoga Mat proves to be a valuable innovation in the field of fitness and rehabilitation. By providing real-time feedback, posture correction, and personalized asana suggestions, it enhances the overall yoga experience. Future improvements could focus on expanding the system's functionality by integrating more advanced sensors for balance analysis, further refining the accuracy and effectiveness of the mat.

CHAPTER - 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Weight-Based Smart Yoga Mat successfully integrates technology with traditional yoga practice, providing real-time weight monitoring and posture correction to enhance user experience. By detecting weight distribution and offering instant feedback, the system helps users maintain balance, improve alignment, and reduce the risk of injuries. It ensures that practitioners perform yoga safely and effectively, making it a valuable tool for both beginners and advanced users. This innovation is particularly beneficial for individuals undergoing physiotherapy and rehabilitation, as it allows them to monitor their progress and track weight variations over time. Unlike conventional yoga mats, this system provides real-time corrective guidance, eliminating the need for constant supervision. The smart mat's durability and energy efficiency make it suitable for long-term use, ensuring reliability and convenience. The LCD display offers a clear and interactive interface, making it accessible for users of all age groups. The project showcases significant potential in fitness and healthcare, helping users develop better posture, flexibility, and strength. It bridges the gap between traditional yoga practice and modern technology, offering a personalized and data-driven approach. By promoting scientific and structured yoga practice, this system can transform the way yoga is performed, making it more effective, precise, and accessible to a broader audience.

5.2 Future scope

The Weight-Based Smart Yoga Mat has several possibilities for future advancements to enhance its functionality. Advanced posture tracking with additional sensors can improve alignment and stability analysis. AI-based personalization can offer customized feedback by analyzing user weight patterns over time. Adding audio and vibration alerts will provide real-time guidance without requiring users to check a display. The mat can also support pre-set yoga routines tailored to individual fitness levels. In healthcare, it can be used for rehabilitation and physiotherapy, assisting patients in balance and recovery training. A lightweight and foldable design will improve portability, making it easier to use in various settings. Enhancing battery life and energy efficiency will allow for longer usage without frequent recharging. Integration with wearable fitness devices can provide a holistic health monitoring system. Multi-user support can enable instructors or therapists to track multiple users simultaneously. Adapting the mat for elderly individuals and those with mobility issues will improve safety and accessibility. Wireless connectivity can allow seamless updates and improved user experience. A waterproof and dust-resistant design will ensure long-term durability. Implementing gesture-based controls can make user interaction more intuitive. Cloud storage integration will enable users to track progress over time. Future developments will make the smart yoga mat a highly efficient tool for yoga, fitness, and rehabilitation, benefiting a wide range of users.

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APPENDIX

```
/*
  ----------------------------------------------------------------------------
  HX711_ADC
  Arduino library for HX711 24-Bit Analog-to-Digital Converter for Weight Scales
  Olav Kallhovd sept2017
  ----------------------------------------------------------------------------
*/

/*
  Settling time (number of samples) and data filtering can be adjusted in the config.h file
  For calibration and storing the calibration value in eeprom, see example file "Calibration.ino"

  The update() function checks for new data and starts the next conversion. In order to acheive maximum effective
  sample rate, update() should be called at least as often as the HX711 sample rate; >10Hz@10SPS, >80Hz@80SPS.
  If you have other time consuming code running (i.e. a graphical LCD), consider calling update() from an interrupt routine,
  see example file "Read_1x_load_cell_interrupt_driven.ino".

  This is an example sketch on how to use this library
*/
```

```
#include<LiquidCrystal_I2C.h>
#include <HX711_ADC.h>
#if defined(ESP8266)|| defined(ESP32) || defined(AVR)
#include <EEPROM.h>
#endif
LiquidCrystal_I2C lcd(0x27,16,2);
//pins:
const int HX711_dout = 4; //mcu > HX711 dout pin
const int HX711_sck = 5; //mcu > HX711 sck pin

//HX711 constructor:
HX711_ADC LoadCell(HX711_dout, HX711_sck);

const int calVal_eepromAdress = 0;
unsigned long t = 0;

void setup() {
  Serial.begin(57600); delay(10);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);lcd.print("Weight Based");
  lcd.setCursor(0,1);lcd.print("Yoga");
  Serial.println();
  Serial.println("Starting...");

  LoadCell.begin();
```

```
  //LoadCell.setReverseOutput(); //uncomment to turn a negative output value to positive
  float calibrationValue; // calibration value (see example file "Calibration.ino")
  calibrationValue = 94696.0; // uncomment this if you want to set the calibration value in the sketch
#if defined(ESP8266)|| defined(ESP32)
  //EEPROM.begin(512); // uncomment this if you use ESP8266/ESP32 and want to fetch the calibration value from eeprom
#endif
  //EEPROM.get(calVal_eepromAdress, calibrationValue); // uncomment this if you want to fetch the calibration value from eeprom

  unsigned long stabilizingtime = 2000; // preciscion right after power-up can be improved by adding a few seconds of stabilizing time
  boolean _tare = true; //set this to false if you don't want tare to be performed in the next step
  LoadCell.start(stabilizingtime, _tare);
  if (LoadCell.getTareTimeoutFlag()) {
    Serial.println("Timeout, check MCU>HX711 wiring and pin designations");
    while (1);
  }
  else {
    LoadCell.setCalFactor(calibrationValue); // set calibration value (float)
    Serial.println("Startup is complete");
  }
```

```
  delay(3000);
  lcd.clear();
}

void loop() {
  static boolean newDataReady = 0;
  const int serialPrintInterval = 0; //increase value to slow down serial print activity

  // check for new data/start next conversion:
  if (LoadCell.update()) newDataReady = true;

  // get smoothed value from the dataset:
  if (newDataReady) {
    if (millis() > t + serialPrintInterval) {
      float i = LoadCell.getData();
      Serial.print("Load_cell output val: ");
      Serial.println(i);
      lcd.clear();
      lcd.setCursor(0,0);lcd.print("Weight:");
      lcd.setCursor(10,0);lcd.print(i*100);lcd.print("g");
      if((i*100) >20 && (i*100)<40)
      {
        lcd.setCursor(0,1);lcd.print("Do:");
        lcd.setCursor(5,1);lcd.print("Vajrasanam");
        Serial.println("Vajrasanam");
      }
```

```
      else if((i*100) >40 && (i*100)<80)
      {
        lcd.setCursor(0,1);lcd.print("Do:");
        lcd.setCursor(5,1);lcd.print("Badmasanam");
        Serial.println("Badmasanam");
      }
      else if((i*100) >80 && (i*100)<100)
      {
        lcd.setCursor(0,1);lcd.print("Do:");
        lcd.setCursor(5,1);lcd.print("Sirasasanam");
        Serial.println("Sirasasanam");
      }
      newDataReady = 0;
      t = millis();
    }
  }
  // receive command from serial terminal, send 't' to initiate tare operation:
  if (Serial.available() > 0) {
    char inByte = Serial.read();
    if (inByte == 't') LoadCell.tareNoDelay();
  }
 // check if last tare operation is complete:
  if (LoadCell.getTareStatus() == true) {
    Serial.println("Tare complete");
  }
delay(500);
}
```